Endogenous Population Scale in the Existing Housing–Fertility Model

Research update

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Abstract

The paper's lifecycle model determines fertility, tenure, and housing choices for a stationary population. This update adds one feedback: births become future entrant households, and those households add to housing demand. A two-generation model shows how fertility and housing clearing jointly determine long-run housing cost and population. We then add the same population closure around the sequential-fertility model used in the August presentation, without changing its household problem. The current estimate has no positive closed-population steady state: cheaper housing raises fertility, but not enough to restore replacement. It does have well-defined open steady states. An exact cohort-accounting diagnostic connects two of them and shows the twenty-year delay between births and adult entry. Starting from a steady state does not, by itself, generate a period of rising population or house prices. Doing so requires a nonstationary initial cohort and household-formation distribution; welfare additionally requires a forward-looking asset-price transition.

1 The mechanism

Children raise the demand for space at the same ages when households have low liquid wealth and limited access to ownership. Large rental units are scarce, and buying a family-sized home requires a down payment. Older households, by contrast, own much of the large-housing stock and may retain it after their children leave because moving is costly and housing is part of the estate they plan to leave. The allocation of housing across generations can therefore raise the effective cost of a child even when the aggregate stock is large.

The feedback has two directions. First, a higher housing cost or a tighter access constraint reduces fertility. Second, children eventually form the next generation of households, so fertility changes future housing demand. The first direction is a household mechanism; the second is a demographic equilibrium condition. Combining them changes both the transition and the long-run incidence of housing policy.

These links have close precedents. Demographic effects on housing demand go back at least to Mankiw and Weil (1989), and recent evidence relates birth-cohort size to housing prices through entry into and exit from homeownership (Francke and Korevaar, 2025). Housing prices and mortgage access also affect childbearing (Lovenheim and Mumford, 2013; Dettling and Kearney, 2014; Hacamo, 2021). Malmberg (2012) studies the two-way relation between cohort housing pressure and fertility, while Lee and Suh (2025) analyze a low-fertility housing transition. Our question is narrower: how does the incidence of housing policy change when tenure constraints and old-age retention affect fertility, and births in turn determine the number of future households?

The current paper already writes a stationary entry condition, while its household computations describe a population normalized to unit size. Let S be the number of adult households in the market. For any housing cost p, the household model tells us four things: the distribution of households, housing demand per household $\bar{h}(p)$, the number of new households required to replace those that exit $E(p)$, and the number produced by children born inside the model $B(p)$. The objects $E(p)$ and $B(p)$ are flows per adult household and model period.

To allow for immigration, let M be the aggregate number of households arriving from outside in the same period and let ρ be the fraction of locally born households that remain in the market. A stationary population and a clearing housing market then require

$$SE(p) = M + \rho SB(p), \qquad S\bar{h}(p) = H^s(p). \tag{1}$$

The first equation says that local children and outside entrants together replace exiting households. The second says that total housing demand equals housing supply. These two equations determine population S and the housing cost p without changing the household problem.

For a stationary comparison, this is the entire population extension. A transition additionally requires the age distribution and the children who have already been born but have not yet formed households. Those objects are introduced below. Outside and locally born entrants are initially assigned the same entry-state distribution as in the current model; allowing their wealth or income distributions to differ is a later extension.

The next section presents the same mechanism with young and old households. Section 3 then maps it into the current paper.

2 Two-generation theory

2.1 Environment

Young households. At date t, there are Y_t young households. Each has income y and chooses consumption c, housing for the adults h, and the number of children n. Preferences are

$$u(c, h, n) = c + \alpha \log h + \theta \log n, \qquad \alpha > 0, \quad \theta > 0. \tag{2}$$

The parameter α governs the demand for adult housing and θ governs the demand for children. Quasi-linearity makes both demands available in closed form. We assume $y > \alpha + \theta$, so consumption is positive at the interior solution.

Child costs and young access. Each child requires $q \geq 0$ units of other goods and $a > 0$ units of housing. One unit of housing costs p. Obtaining family space also entails an access cost $\omega \geq 0$, which represents a small rental market, a down-payment constraint, or another tenure friction. The cost of a child is therefore $q + a(p + \omega)$.

Old households. There are O_t old households. They no longer choose fertility, and their housing demand is

$$h_O(p) = \frac{\bar{h}_O}{p}, \qquad \bar{h}_O > 0. \tag{3}$$

The parameter $\bar{h}_O$ summarizes old households' desire to retain housing. A policy that encourages downsizing reduces $\bar{h}_O$. In the full model, housing retention follows from moving costs and the value of leaving a house to one's children rather than from this demand schedule.

Housing. Aggregate housing services are supplied according to

$$H^s(p) = \bar{H} p^{\varepsilon}, \qquad \bar{H} > 0, \quad \varepsilon \geq 0. \tag{4}$$

The fixed-stock case is $\varepsilon = 0$.

Demography. The number $\bar{n} > 0$ is replacement fertility. After survival and household formation, $\bar{n}$ children produce one new young household.

In this static section, p is the current cost of one unit of housing services. The lifecycle model distinguishes that current rent, denoted r_t, from the asset price P_t that capitalizes future rents. They move proportionally only under a constant user cost and a stationary price path.

2.2 The young household

Taking p as given, a young household solves

$$\max_{c,h,n}\ c+\alpha\log h+\theta\log n \quad \text{subject to} \quad c+ph+[q+a(p+\omega)]n=y. \tag{5}$$

At an interior solution, the household chooses

$$n(p)=\frac{\theta}{q+a(p+\omega)}, \qquad h_Y(p)=\frac{\alpha}{p}+an(p). \tag{6}$$

Thus a higher housing cost or a tighter access constraint reduces fertility. Young housing demand is adult housing, α/p, plus the space required by children, $an(p)$.

This is a stripped-down version of the household mechanism already in the paper. There, the other-goods cost of children is χ, the space required per child is κ, and the relevant housing cost depends on tenure and on whether the household is constrained. In the notation of the current model, the simple expression $q+a(p+\omega)$ corresponds to $\chi+\kappa(r^m+\zeta_i^m)$: ω summarizes the shadow cost of limited housing access. The simple model removes heterogeneity, saving, and the rent-versus-own decision only to make the demographic feedback transparent.

2.3 Equilibrium and the steady state

Housing clearing adds young and old demand:

$$\bar{H}p_t^{\varepsilon}=Y_t h_Y(p_t)+O_t h_O(p_t). \tag{7}$$

For inherited cohort masses, the left side rises and the right side falls with p_t, so the clearing price is unique.

Children become the next young cohort, while the current young become old:

$$Y_{t+1}=\frac{n(p_t)}{\bar{n}}Y_t, \qquad O_{t+1}=Y_t. \tag{8}$$

The ratio $n(p_t)/\bar{n}$ is the number of next-period young households produced by each current young household.

A steady state has a constant housing cost and constant positive cohort masses, so $n(p^*)=\bar{n}$ and $Y^*=O^*$.

Proposition 1 (Steady state)**.** *A unique positive steady state exists if and only if $\theta>\bar{n}(q+a\omega)$. It is*

$$p^*=\frac{\theta/\bar{n}-q}{a}-\omega, \qquad Y^*=O^*=\frac{\bar{H}(p^*)^{\varepsilon+1}}{\alpha+\bar{h}_O+a\bar{n}p^*}. \tag{9}$$

A lower value of children θ or a higher access cost ω lowers both the steady-state housing cost and population. A higher $\bar{h}_O$ reduces population but does not change the steady-state housing cost. A proportional increase in housing supply $\bar{H}$ raises population in the same proportion and also leaves the steady-state housing cost unchanged.

The cohort law first determines the housing cost consistent with replacement, $n(p^*)=\bar{n}$. Housing clearing then determines how many households can live in the market at that cost. This is why a permanent increase in housing supply raises population rather than lowering the steady-state housing cost.

Even if market housing were free, a child would still cost $q+a\omega$. Hence cheaper housing cannot restore replacement when $\theta\le\bar{n}(q+a\omega)$.

2.4 Comparative statics

Suppose the economy begins in a steady state and a permanent reform arrives after Y_t and O_t have been inherited. The impact and steady-state effects are:

	Impact p_t	Impact n_t	Long-run p^*	Long-run n^*	Long-run population
Lower access cost, $\omega \downarrow$	+	+	+	0	+
Lower old housing demand, $\bar{h}_O \downarrow$	−	+	0	0	+
Greater housing supply, $\bar{H} \uparrow$	−	+	0	0	+

Lowering ω makes family housing easier to obtain. Fertility and housing demand rise immediately, so part of the reform appears as a higher market price. Lower old housing demand and greater supply instead reduce the price on impact. In the new steady state, fertility is again $\bar{n}$. For the latter two reforms, the housing cost returns to its original level and the additional housing supports a larger population.

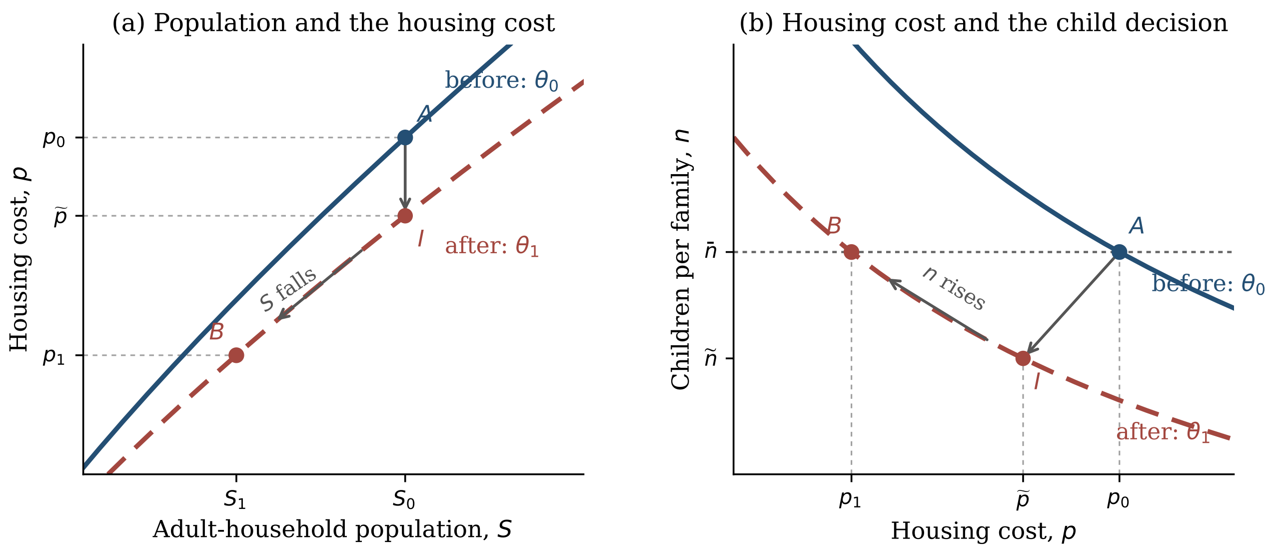

Figure 1: The child decision determines the steady-state housing cost through $n(p^*) = \bar{n}$. Housing clearing then determines population at that cost. A fall in θ initially puts fertility below replacement; as population falls, housing becomes cheaper and partly offsets the initial decline. The solid blue housing-clearing schedule applies before the fall in θ and the dashed red schedule after it; A, I, and B denote the initial steady state, impact allocation, and new steady state.

2.5 Demographic momentum

Let $\mu_t = O_t/Y_t$ be the number of old households per young household, and let $m_t = n(p_t)/\bar{n}$ be the number of next-period young households produced by each current young household. When fertility is below replacement, $m_t < 1$, total household population still rises if

$$\mu_t < m_t < 1. \tag{10}$$

For example, 100 young and 70 old households become 80 young and 100 old households when $m_t = 0.8$. Population rises from 170 to 180 even though the young cohort is not replacing itself.

If preferences and housing supply do not change again between t and $t+1$, the housing cost also rises when

$$(1 - \mu_t)h_O(p_t) > (1 - m_t)h_Y(p_t). \tag{11}$$

The smaller young cohort reduces demand by $(1-m_t)Y_t h_Y(p_t)$. At the same time, the current young cohort replaces a smaller old cohort and raises old demand by $(1-\mu_t)Y_t h_O(p_t)$. The price rises when the second change is larger.

Starting from an exact steady state, $\mu_t = 1$. In this closed two-cohort model, a fertility decline lowers the housing cost on impact and lowers population in the next period. A period in which fertility is below replacement while population and housing costs rise therefore requires an inherited cohort imbalance, produced by earlier fertility, immigration, or both.

On impact, the fertility decline lowers the housing cost relative to the no-shock allocation. Demographic momentum can nevertheless make the realized cost rise over time. The sign comparison above does not by itself order the entire shocked and no-shock price paths.

2.6 Intergenerational policy

Lowering ω directly improves young households' access to family space. Lowering $\bar{h}_O$ instead releases housing from old households and reduces the market price during the transition. Both policies raise fertility, but through different margins. A general expansion of supply also lowers the transition price, without changing which generation initially releases the housing.

With a fixed adult population, reallocating a home toward a constrained young household affects current tenure and fertility. With the cohort law in equation (8), the resulting birth also changes the number of future entrants. That second effect is lost when population is renormalized after every solution.

3 Quantitative implementation

3.1 Minimal wrapper around the sequential model

The quantitative model is the one-market sequential-fertility specification used in the August presentation. Households make age-dependent first- and later-birth decisions, children ever born are distinct from children currently at home, and the model retains income risk, saving, tenure choice, the owner ladder, and bequests. The population closure changes none of those choices.

At a candidate asset price P, the stationary solver returns the adult entry flow $E(P)$, effective locally born entrant households $B(P)$, and housing demand per adult household $D(P)$. For a fixed outside entrant flow M and local retention ρ, stationary adult-household population is:

$$S(P) = \frac{M}{E(P) - \rho B(P)}. \tag{12}$$

The denominator is the entry shortfall left after retained local children. An open steady state is finite when it is positive. Housing clearing then chooses P from:

$$S(P)D(P) = H^s(P). \tag{13}$$

Thus the stationary extension is an outer calculation around the existing solver; it introduces no new household state.

For comparison, a positive closed steady state would require both replacement and housing clearing:

$$B(P^*) = E(P^*), \qquad S^* = \frac{H^s(P^*)}{D(P^*)}. \tag{14}$$

The first condition need not have a solution at the estimated household parameters. This is a substantive restriction, not a numerical normalization.

3.2 The current estimate has no closed steady state

The calculation holds the working parameter estimate fixed. That estimate is not yet a final calibration: it misses childlessness, mean occupied rooms, and old-age wealth dispersion, and its first-birth housing target remains under empirical review.

Under the fiscally balanced 1 percent property-tax convention, required entrants are $E_0 = 0.061733$ per adult household and effective locally born entrants are $B_0 = 0.052929$. The resulting replacement ratio is:

$$\frac{B_0}{E_0} = 0.857387. \tag{15}$$

The calibrated fertility response has the expected sign but is small. Reducing the asset price to one half of one percent of its baseline value raises $B(P)/E(P)$ only to 0.874582; increasing the price to three times baseline reduces it to 0.818661. The schedule is monotone over this range and never crosses one. Equation (14) therefore has no solution at the current parameters.

Figure 2 shows the two stationary objects. The left panel is the replacement condition; the right panel records the population scale allowed by housing clearing at each price. The distance from the horizontal replacement line is the main quantitative finding: the present model supports an open-population endpoint, not a closed one.

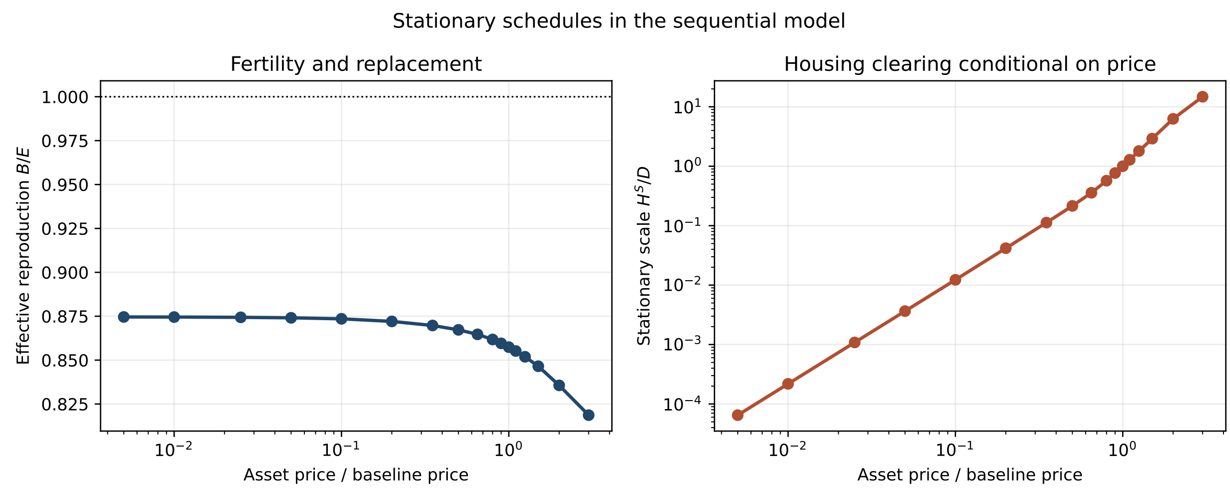

Figure 2: Stationary schedules in the sequential model. The left panel plots effective locally born entrant households relative to required entrants. The right panel plots the scale permitted by the maintained housing-supply schedule, conditional on price. The working estimate remains below replacement throughout the displayed range.

3.3 How the closure changes the paper's policy experiments

The standing fiscally balanced policy cases are also run under the sequential model. The reference is a 1 percent annual property tax on occupied housing, rebated equally. The first counterfactual raises the rate to 2 percent. The second also provides the existing purchase grant for a dependent-child renter who buys a home with at least six rooms. The parameter estimate is fixed, so these are mechanism calculations rather than final policy estimates.

Outside entrants are 16.9 percent of baseline entrants. Local retention is chosen to reproduce the reference population exactly, and both objects are then held fixed. Total births combine births per adult household with the change in adult-household population.

The price effect remains the main response, as in the fixed-population exercise. Endogenous population adds a quantitatively smaller but nontrivial margin: lower housing costs raise fertility per household,

Table 1: Policy effects under the open stationary closure

	ΔP	ΔS	Δ births/adult	Δ total births
Rebated 2% property tax	-17.53	$+1.69$	$+0.33$	$+2.03$
Rebated 2% tax + purchase grant	-17.29	$+3.46$	$+0.68$	$+4.16$

Notes: Percent changes from the fiscally balanced 1 percent property-tax reference. Population is adult-household mass.

and the resulting steady state contains more households. The magnitudes depend on the outside share and retention rate, which are provisional. These rows establish the endpoints; they do not describe the path between them or welfare.

3.4 Movement between open steady states

The calendar wrapper carries the full adult distribution over age, liquid wealth, income, tenure, parity, and children at home. Its one-period operator reproduces the stationary distribution with L1 error 8.04×10^{-17} and the effective locally born entrant flow with error 1.57×10^{-15}. Population mass is never renormalized.

The household child state is deliberately not used as the demographic clock. Although it tracks each child's exit independently, its geometric transition allows a newborn to mature in the same four-year period. Population renewal therefore carries a separate aggregate birth-vintage queue. Births affect housing choices immediately, but affect adult entry twenty years later. This adds timing to the closure without changing any household choice.

For a transparent diagnostic, set the child-preference shifter to 0.10 in the old steady state and permanently reduce it to its estimated value of zero. This path uses the calibration convention without a contemporaneous property tax rebate; the fiscally balanced rows above are separate stationary policy comparisons. At the old point, TFR is 2.111, $B/E = 0.9075$, and the asset price is 0.7594. An outside share of 16.9 percent implies retention $\rho = 0.9157$. Holding the outside flow, retention, and housing stock fixed, the new steady state has:

$$\frac{S^{\text{new}}}{S^{\text{old}}} = 0.7936, \qquad \frac{P^{\text{new}}}{P^{\text{old}}} = 0.7304, \qquad \text{TFR}^{\text{new}} = 1.985. \tag{16}$$

The preference change therefore connects two positive open steady states. The fall in population lowers the price, and that lower price partly offsets the fertility decline, but it does not restore replacement.

After 96 years, adult-household mass is 0.9110 and the house-price index is 0.8740, so convergence is slow. The transition does not generate an initial boom: from an exact old steady state, population is flat during the delay and the price edges down. A period of rising population and prices is possible with inherited cohort momentum, as the simple model shows, but it must be in the initial distribution; it does not follow from below-replacement fertility alone.

3.5 What remains before calling this the U.S. transition

The 2023 age distribution cannot yet be inserted consistently. In the ACS large-metro sample, ages 18–21 are 0.94 percent of household heads aged 18–85, while the model injects 6.17 percent of adult-household mass into its age-18 entrant state. Joining them creates an artificial boom that is really household formation. Reweighting by all adults reduces that jump but attaches household wealth, tenure, and housing demand to persons rather than households.

The empirical bridge is therefore age-specific household formation. A final welfare transition also requires households to anticipate the path of rents and resale prices. The calculation above instead treats the current price and post-shock primitives as permanent at each date. It is useful for cohort accounting and for deciding whether the mechanism has the right timing, but it is not a U.S. forecast or a perfect-foresight asset-price equilibrium.

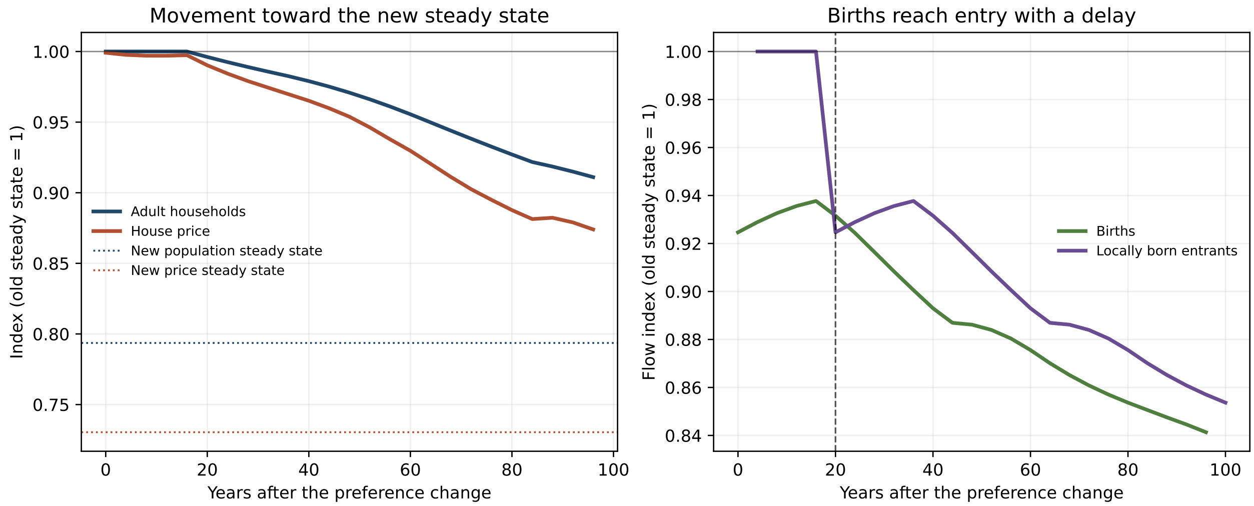

Figure 3: Temporary-equilibrium movement between open steady states. Births respond when preferences change, while locally born entry remains at its old flow until the birth-vintage queue turns over. Adult-household population is therefore unchanged for twenty years and then declines. Dotted lines mark the new stationary endpoints.

Finally, the long-run strength of the feedback depends on the causal response of fertility to housing costs and access. Fertility levels at one price do not identify that slope. The current no-root result should therefore be revisited after the fertility and first-birth housing targets are repaired and the model is refit.

4 Conclusion

The proposed change is narrow: an outer loop uses the existing model's entrant, child-maturation, and housing-demand objects to determine population and the housing price jointly. The sequential household problem and the paper's intergenerational mechanism remain intact.

At the current parameter estimate, cheaper housing raises fertility but never restores replacement, so the relevant endpoints are open steady states. The calendar wrapper connects two such endpoints and delivers the twenty-year lag from births to new adult households. Starting from a steady state, however, it does not reproduce the recent combination of rising U.S. population and house prices. The next empirical step is therefore to measure age-specific household formation and use it to construct a nonstationary initial distribution. A perfect-foresight price path is needed only for the subsequent welfare exercise.

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